

Your own kubernetes lab with k3s



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<https://github.com/gdha/k3s-intro>

What is kubernetes?



- **K**ubernetes = K8s
- Container orchestration platform
- Network, DNS, security, load balancers, storage drivers all included
- High Availability out of the box
- Container = application isolation
- Upgrade K8s on-line without interruption of running containers

Container and Kubernetes are the defacto standard



- Containers are awesome
 - Consistent across a wide variety of infrastructure
 - Capable of standalone or clustered operations
 - Easy to upgrade and/or replace containers
 - Support for different infrastructure config (storage, cpu, ...)
 - Strong ecosystem (monitoring, logging, CI, management, ...)
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Kubernetes on the edge challenges



- Most Kubernetes distributions don't support ARM
- Kubernetes could easily consume up to 4gb of RAM
- Kubernetes wasn't built for embedded or offline management
- Need for simplified operations



Kubernetes variations



- EKS from AWS (Amazon)
 - GKE from GCP (Google)
 - AKS from Azure (Microsoft)
 - RKE from Rancher
 - PKS from Pivotal (VMware)
 - DKS from Docker
 - LKE from Linode
 - OKD and/or OpenShift (RedHat)
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Introducing k3s



- Lightweight certified Kubernetes distro
- Built for production operations
- 40MB binary, 250MB memory consumption
- Single process w/ integrated Kubernetes master, Kubelet, and containerd
- SQLite in addition to etcd
- Available for x86_64, ARM64, and ARMv7
- Open source project (not a Rancher product yet)

k3s = k8s - 5



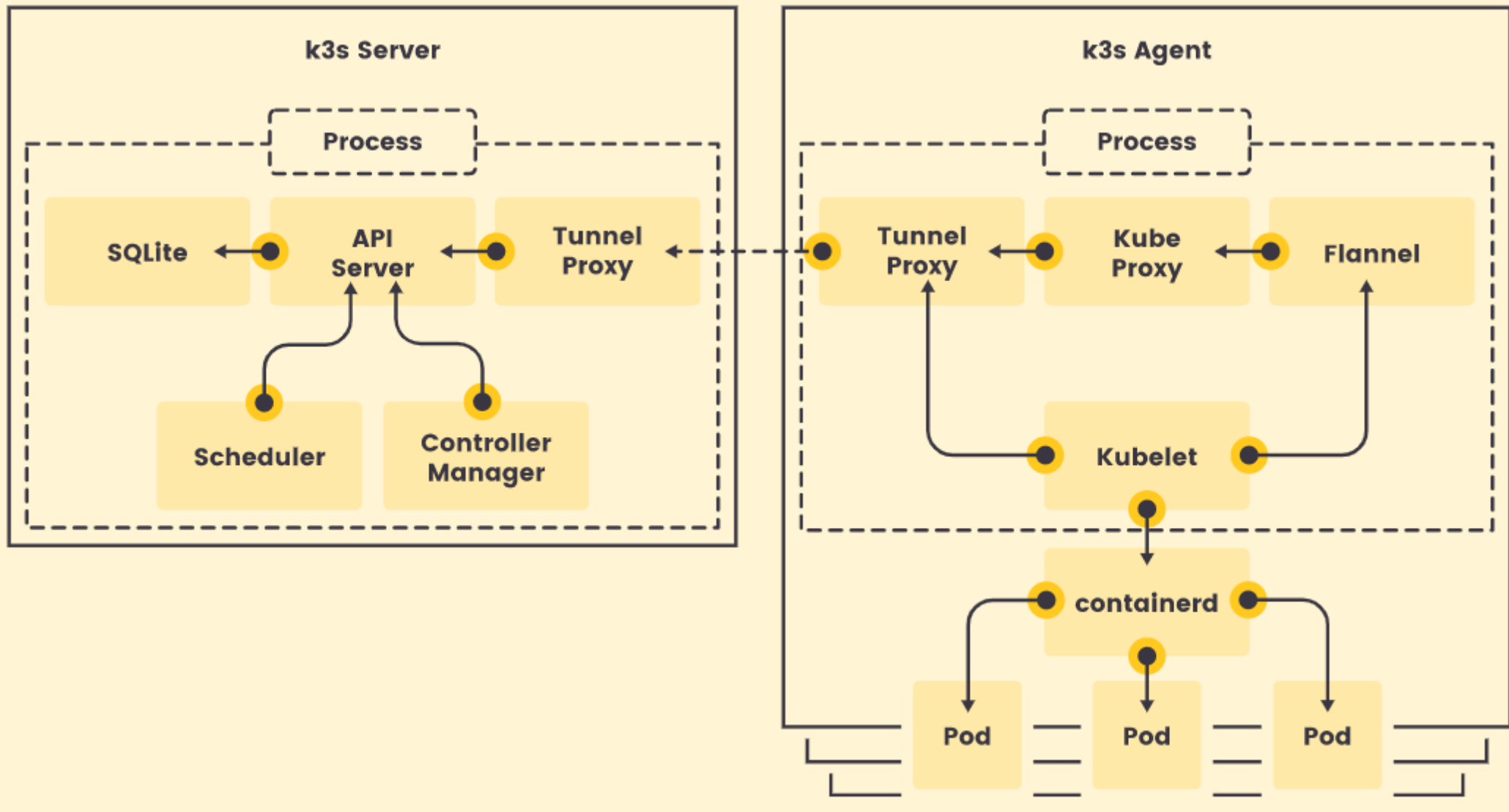
Removes

- Legacy and non-default features
- Alpha features
- Cloud providers
- 3th party storage drivers
- Docker (optional)

Adds

- Simplified installation
 - SQLite3 support
 - TLS management
 - Automatic Manifest
 - Helm Chart management
 - Containerd, CoreDNS, Flannel
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How k3s works



K3s pointers



- Home: <https://k3s.io/>
- Sources: <https://github.com/rancher/k3s>
- Docs: <https://rancher.com/docs/k3s/latest/en/>
- Rancher Lab youtube video channel:
<https://www.youtube.com/channel/UCh5Xtp82q8wjijP8npkVTBA>



Educational



<https://github.com/vmware-tanzu/tgik/blob/master/playlist.md>



TGI
kubernetes

Demo: lab k3s cluster



<https://github.com/gdha/k3s-intro>

Ubuntu 18.04.3 LTS vagrant boxes

K3s installed and configured with **ansible**

Node1
(master)

192.168.33.11

Node2
(worker)

192.168.33.12

Node2
(worker)

192.168.33.13

Demo: initial k3s cluster



- vagrant up --provision
- Login into node1
 - ssh vagrant@192.168.33.11
 - vagrant ssh node1

```
vagrant@node1:~$ kubectl get pods -A
```

NAMESPACE	NAME	READY	STATUS	RESTARTS	AGE
kube-system	local-path-provisioner-58fb86bdfd-ddpzk	1/1	Running	0	34m
kube-system	metrics-server-6d684c7b5-nddkt	1/1	Running	0	34m
kube-system	tiller-deploy-6966cf57d8-m255c	1/1	Running	0	33m
kube-system	coredns-d798c9dd-h8bqk	1/1	Running	0	34m
kube-system	helm-install-traefik-5n4fz	0/1	Completed	0	34m
kube-system	svclb-traefik-8w4tx	3/3	Running	0	32m
cert-manager	cert-manager-cainjector-6659d6844d-8ft75	1/1	Running	0	32m
cert-manager	cert-manager-6bcdcf8c5cc-skp4n	1/1	Running	0	32m
kube-system	traefik-65bccdc4bd-24dtt	1/1	Running	0	32m
cert-manager	cert-manager-webhook-547567b88f-n75ww	1/1	Running	0	32m
kube-system	svclb-traefik-6nq88	3/3	Running	0	26m
kube-system	svclb-traefik-slwtm	3/3	Running	0	21m

- [illegible]